Thresholds That Travel: Adaptive Alarm Thresholds Let Attacks Through in ICS Anomaly Detection

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Abstract

Anomaly detectors for industrial control systems (ICS) raise an alarm when a score crosses a threshold. The threshold is calibrated on clean data and must then hold a false-alarm budget for months, so deployments often let it adapt: they recalibrate it daily, track a running quantile, or refit an extreme-value tail as data arrive. Each rule learns from part of its input, and we ask what happens when attacks land there.

In three pre-registered studies on data that no earlier analysis of ours had opened, a symmetric quantile tracker lost attacks that a frozen threshold caught: 20 of HAI 20.07's 38 against 36 (difference -0.42, 95% interval $[-0.58, -0.26]$), 6 of BATADAL's 14 against 14 ($[-0.79, -0.29]$) and, on the Tennessee Eastman process, 66 of the 104 attacks that follow another attack in their run against 104 ($[-0.47, -0.27]$). On the first attack of a run the two were equal (116 of 141 each). A tracker calibrated to HAI 20.07's false-alarm rate needs 23 exceedance-free hours to undo one minute above its threshold, whatever its step size. Under a stricter count, in which an alarm must begin inside the attack (registered on Tennessee Eastman, post-hoc on the other two), the loss holds on BATADAL but not on HAI 20.07 or Tennessee Eastman, where later attacks arrive inside alarms already running; that failed the third study's primary rule. On a fourth dataset, ProvICS, whose monitor was in alarm for 44% of the unlabelled time, the tracker lost nothing.

Deliberate poisoning is cheaper. On 56 sealed attack-free Tennessee Eastman runs (registered), a day of two-minute nuisance alarms left the tracker catching none of 56 later, just-detectable attacks; pulses shorter than what a rule counts as an event did the same to a recalibration that leaves alarm episodes out; and an attacker who held the score just under the current threshold left a standard SPOT threshold catching 16 of 56 attacks a quarter above the calibrated threshold. The frozen threshold caught all 56 each time and ran at 0.98 times its false-alarm budget over 280 attack-free days. A threshold that only tightens itself cannot miss an event the frozen threshold catches, and a ceiling at κ times the calibrated threshold confines every one of these attacks to events at or below the ceiling. We argue that an alarm threshold may tighten itself but should loosen only inside a fixed band or with operator approval, and we release the registrations, ledger and code.

Keywords

industrial control systems, anomaly detection, adaptive thresholds, poisoning, extreme value theory, pre-registration

1 Introduction

A process-monitoring anomaly detector has two parts: a score that grows when the plant behaves unlike its normal operation, and a threshold that turns the score into alarms. Most of the research effort goes into the score. The threshold is usually set once, on held-out normal data, to meet a false-alarm budget, and then expected to hold that budget on a plant that drifts: sensors age, seasons change load, and operators retune loops. When it does not hold, the standard remedies adapt the threshold online. Daily recalibration re-estimates a quantile on the last day of data. Online quantile tracking, the mechanism behind adaptive conformal inference [15], nudges the threshold up after every exceedance and down otherwise. Streaming peaks-over-threshold methods (SPOT and DSPOT [31]) refit a generalised Pareto tail as new values arrive, and the nonparametric dynamic threshold of Telemanom [19] recomputes its level on a trailing window.

Each of these rules learns from part of its input. Trackers and recalibration learn from the values that exceed the threshold. SPOT and DSPOT discard values above their alarm level but learn from values between a base level and the alarm level. A frozen threshold learns from nothing after calibration. We call the part of the score range a rule learns from its *learning region*, and the question of this paper is what happens when an attack, or an attacker, lands in it.

The answer depends on the region, and none of the rules we tested that can raise their own threshold is safe:

- **Rules that learn from exceedances are poisoned by real attacks.** An attack is a long exceedance, so a tracker raises its threshold during the attack and stays high afterwards. The next attack then arrives under an inflated threshold. On fresh HAI 20.07 data the tracker missed 18 of 38 attacks, 16 of them attacks the frozen threshold caught; on BATADAL it missed 8 of 14, all caught by the frozen threshold; on Tennessee Eastman it missed 38 of the 104 attacks that follow another attack and no more than the frozen threshold among first attacks. No adversarial planning is needed; the attacks in the recording did it.
- **The same rules can be poisoned on purpose and loudly.** Twelve two-minute pulses a day, the kind of nuisance alarm operators learn to acknowledge, raised these thresholds enough to hide a later attack that a frozen threshold caught every time (registered on sealed attack-free Tennessee Eastman runs; exploratory on HAI).
- **Rules that refuse to learn from alarms can be poisoned silently.** An attacker who reads the current SPOT or DSPOT threshold and holds the score just under it feeds every sample into the fitted tail. The tail extends, the threshold rises, and the attacker follows it up. At the score level the campaign usually raises no alarm (none in 51 of the 56 registered placements). A standard SPOT then caught 16 of 56 later attacks a quarter above the calibrated threshold on the sealed Tennessee Eastman runs, against 56 without the campaign (registered).

- **Rules that leave alarms out of their update can be poisoned below their notice.** A recalibration that excludes alarm episodes must decide what an episode is. Pulses shorter than its event length do not count, so it learns from them (registered).

A threshold that only tightens itself cannot be pushed above the frozen threshold by any of these, so it keeps the frozen threshold's recall by construction (Section 3.1). Its cost is extra false alarms. A ceiling at $\kappa\tau_0$ extends the guarantee to rules that loosen: no poisoning can hide an event above the ceiling. Loosening beyond it is left to an operator, and that step carries no such guarantee.

Method. Three studies were registered, each before any test row or label of its datasets was read: tt1 (HAI 20.07, HyGrid-ICS, vorausAD), tt1b (BATADAL) and tt1d (Tennessee Eastman, ProvICS; for these the dataset documentation, the file names and the header lines of the label files were read first). A ledger records every access to recorded attack data, the analysis harness refuses to run if any frozen source file changed after registration, and it computes the registered gates itself. The rules and the attack designs came from development on three earlier HAI releases whose attacks had already been used; we report how the claim changed during development, including two hypotheses we dropped. Extensions run after the registered finals are labelled exploratory.

Contributions.

(1) Three registered studies of threshold rules on six never-opened datasets (hardware-in-the-loop and simulated plants, an industrial robot), reporting every registered outcome, including three failed primary rules and one dated amendment (Sections 5 and 6).
(2) Evidence that exceedance-learning thresholds lose recall on real attacks because earlier attacks inflate them: registered on HAI 20.07, BATADAL and Tennessee Eastman, where the tracker's loss falls on attacks that follow attacks and not on the first attack of a run. Under a stricter onset count (registered on Tennessee Eastman, post-hoc elsewhere) the loss survives on BATADAL only, and ProvICS shows none.
(3) A learning-region analysis of adaptive thresholds: how long the tracker and daily recalibration remember an exceedance (Section 3.1), and a matching attack for each class, registered on sealed attack-free runs: loud pulses for exceedance learners, a closed-loop frog-boiling attack for extreme-value learners, and sub-event pulses for rules that leave alarms out (Section 6).
(4) A design rule, supported by these results: thresholds may tighten themselves; loosening needs a ceiling or a human (Section 8).
(5) The registration, access ledger, harness and verifier, in the public repository at https://github.com/VyomKulshrestha/Ferrum-OS.

2 Background and related work

ICS anomaly detection. Process-level detectors learn normal behaviour from sensor and actuator traces and flag deviations: model residuals, autoencoder reconstruction errors, or statistical bounds. Public testbed recordings with labelled attacks (SWaT [16], WADI [1], BATADAL [33], and the HAI releases: HAI 1.0 (20.07) [29], 21.03 [30], later releases in the dataset repository) are the usual benchmarks. Evaluations usually fix one threshold per detector, chosen on clean data, by an offline peaks-over-threshold fit [32], or, less defensibly, on the test labels. Wu and Keogh [36] list further benchmark pitfalls.

Threshold rules. Static thresholds meet a quantile or false-alarm budget on clean data. Adaptive variants recalibrate on a trailing window, track a target exceedance rate online [15], fit an extreme-value tail and update it per observation (SPOT, and DSPOT for drifting streams [31]), or recompute a level on a trailing window of smoothed errors [19]. Game-theoretic time-varying thresholds [14] are precomputed, not learned from the stream. Alarm-management practice treats nuisance and chattering alarms as a condition to eliminate, with explicit alarm-rate targets, and routes changes to an alarm through authorised management of change [11, 20]; operators who acknowledge nuisance alarms without investigating them are the failure mode our pulse campaigns exploit.

Poisoning and stealthy attacks. Rubinstein et al. [26] showed that slowly growing ("boiling frog") poisoning of a weekly retrained PCA detector, whose threshold is re-estimated with it, can stop it flagging a later attack. Kloft and Laskov [21, 22] showed that bounding the false-positive rate limits poisoning of online centroid detectors. Chan-Tin et al. [6, 7] described frog-boiling against outlier rejection in network coordinate systems: every update stays within the rejection threshold while the system drifts. In ICS, Kravchik and Shabtai [24] found an autoencoder on SWaT physical-state data resistant to poisoning, but later work poisoned online-retrained ICS detectors with sub-threshold samples and proposed lowering the threshold as mitigation [23], and Seo et al. [27] poisoned a drift-adaptive SWaT detector through its retraining step, evading its drift detector until the retrained threshold cut precision from 0.91 to 0.24. Erba et al. [12] concealed attacks from reconstruction-based detectors by manipulating sensor readings. Attackers who hold a detection statistic just under a fixed threshold are standard in control security [5, 34]. Hong [17] studied gradual drift against world-model monitors in simulation, without attacks or adaptive thresholds. These works target the model or the score, or assume a fixed threshold. We target the adaptive threshold rule and show that it can fail even when the model is not poisoned.

Distribution shift and calibration. Adaptive conformal inference and its successors keep a long-run miscoverage rate under arbitrary, even adversarial, sequences [3, 15], a guarantee about the rate of exceedances. Atarmla [2] proves that for the online quantile tracker this guarantee is an accounting identity and bounds the evidence a step or ramp produces before the tracker absorbs it. The guarantee holds whether the exceedances come from benign drift or from attacks, the property we exploit in Section 7; Proposition 1 (Section 3.1) follows from the same identity, which we test on recorded attacks, and we extend the question to extreme-value thresholds.

3 Threat model and learning regions

System. A monitor computes a score s_t from plant telemetry and raises an alarm when $s_t > \tau_t$. The rule that sets τ_t is calibrated once on clean data to a false-alarm budget and then runs online. Operators acknowledge alarms; persistent nuisance alarms are acknowledged without investigation.

Learning region. For a rule R, the learning region $L(R, \tau_t)$ is the set of score values that change R's future thresholds. Table 1 gives the rules we study.

A rule can loosen (raise τ_t) only through values in the part of its learning region that pushes upward. That gives three attacker classes, from weakest to strongest:

- **A0, no attacker.** Real attacks are themselves exceedances. A rule whose upward learning region includes exceedances is poisoned by the attacks it is meant to detect.
- **A1, open-loop injection.** The attacker can perturb one control loop (setpoints or sensor readings) without seeing the monitor. Supra-threshold pulses land in the exceedance region; sub-alarm pulses land in the band.
- **A2, closed-loop.** The stealthy attacker of control security [5, 34], turned against an adaptive threshold: it can read τ_t (white-box, or by probing) and shape the score. Holding s_t at $0.97\,\tau_t$ places every sample at the top of the band with no alarm. We evaluate A2 at the score level, an upper bound on what a physical attacker can do, and through noise spoofed onto three sensor readings of one loop, scored by the real monitor.

The attacker's goal is a terminal attack that the frozen threshold would detect but the adapted threshold does not.

Why poison rather than conceal. An A2 attacker who can spoof sensors could also try to hide the terminal attack directly, by spoofing the readings it disturbs. Poisoning moves that work earlier: after the campaign the terminal step runs unmodified, so the attacker does not have to keep every reading it changes consistent with normal operation, which constrained-concealment attacks must do [12]. We did not compare the two strategies. The A1 assumption that operators acknowledge recurring alarms without investigating is the condition alarm-management standards try to remove [20]; our pulse campaign adds twelve pulses a day, and the frozen threshold alarmed on 92% of them on HAI 20.07 (registered E4).

3.1 How long a threshold remembers

Three facts follow from the update rules alone. They predict most of what Sections 5 and 7 measure. Proposition 1 restates, as a memory, the counting identity behind the long-run coverage guarantee of adaptive conformal inference [15], from which Atarmla [2] derives the excess alarms a persistent step produces before the tracker absorbs it. What we add is its reading in terms of the calibrated false-alarm rate and its check on recorded attacks.

Proposition 1 (tracker memory). Under $q_{t+1} = q_t + \eta(\mathbf{1}[s_t > q_t] - p)$, each step with $s_t > q_t$ raises the threshold by $\eta(1-p)$ and each other step lowers it by ηp. A rise built from E exceedance steps therefore takes $E(1-p)/p$ quiet steps to undo, whatever the step size η.

The calibrated exceedance rate p sets the memory. On HAI 20.07 ($p = 7.2 \times 10^{-4}$ per second) the ratio is 1,385, so one minute above the threshold keeps it raised for 23 hours. On BATADAL ($p = 0.0075$ per 15-minute step) it is 133, so one step above the threshold keeps it raised for 33 hours. The step size sets how high an exceedance lifts the tracker, not how long the lift lasts: on BATADAL, where η is $6.9\,\tau_0$ per step, one exceedance step lifts it by almost seven times τ_0. A slower tracker (tracker_slow, $\eta/10$) climbs less during a short attack but needs the same 1,385 quiet seconds per exceedance second to come back down, which is why it lost recall too (Section 5.2). Recovery time bounds blind time from above: the tracker misses a later attack only while it sits above that attack's score. Post-hoc checks on the recorded trajectories agree with the proposition: every change of the tracker on the HAI 20.07 and BATADAL test streams is a whole number of these up and down steps (largest deviation 7×10^{-6} of a step), and from its final levels the tracker would have needed 43 to 209 days of exceedance-free data to return to τ_0.

Proposition 2 (daily memory). Daily recalibration sets the next day's threshold to the $(1-p)$ quantile of the last day's N steps. An attack that holds the score above a level a for more than $pN+1$ steps of one day lifts the next day's threshold to at least a.

That is 64 seconds on HAI 20.07 and two 15-minute steps on BATADAL. The memory is one day, so daily recalibration is poisoned whenever attacks fall on consecutive days, as they do in the HAI test files, and it still catches an attack whose first day follows a quiet one. On BATADAL, whose attacks last one to five days and begin 1.7 to 24 days after the previous one ends (post-hoc, from the recorded labels), it caught 14 of 14 with its threshold at $0.69\,\tau_0$ at the median onset (registered tt1b).

Proposition 3 (tighten-only). If $\tau_t \le \tau_0$ at every step (tighten-only without a loosening permit), every sample that alarms under the frozen threshold also alarms under τ_t. The rule therefore detects every event the frozen threshold detects when any alarm inside the event counts, on any input and against any attacker, adaptive ones included. What it can lose is false alarms. Under the stricter onset count of Section 5.2 the guarantee need not hold, because a lower threshold can start an alarm episode before an attack begins.

Proposition 4 (ceiling). If $\tau_t \le \kappa\tau_0$ at every step, every sample above $\kappa\tau_0$ alarms, so under the any-alarm count the rule detects every event that the static threshold $\kappa\tau_0$ detects, on any input and against any attacker. Proposition 3 is the case $\kappa = 1$. A ceiling bounds poisoning: whatever the stream held before, an attacker can hide only events whose score never exceeds $\kappa\tau_0$.

Proposition 3 means the tighten-only results below measure its cost, not its recall. On HAI 20.07, HyGrid-ICS and voraus-AD its recall and false alarms equalled the frozen threshold's: the inflated tracker under the cap kept it at τ_0 (post-hoc: on HAI 20.07 it sat below τ_0 in 0.4% of minutes). Its own behaviour shows on BATADAL, where the tracker sat below τ_0 for 3.2% of the test time (post-hoc) and the rule ran at 2.43 times the false-alarm budget against 0.70 for frozen (registered), and on a development split of HAI 23.05 (5.98 against 4.05, exploratory). A loosening permit breaks the guarantee, even one that only an operator could grant: with the registered oracle permit, a statistical permit that a simulated operator

Table 1: Update and learning region of each threshold rule.

Rule	Update	Learning region
frozen, static margin	none	empty
symmetric tracker $q_{t+1} = q_t + \eta(\mathbf{1}[s_t > q_t] - p)$	up on exceedance, down otherwise	all values; upward moves only from $s_t > \tau_t$
daily recalibration	$(1-p)$ quantile of the last day	the top p of each day, whatever its source
SPOT	adds $s_t - t$ to the tail fit when $t < s_t \le z_q$	the band $(t, \tau_t]$
DSPOT	as SPOT on s_t minus a moving mean of non-alarm values	the band, plus every non-alarm value through the mean
tighten-only	min(ceiling, tracker); the ceiling rises only through a permit	downward only, unless a permit fires

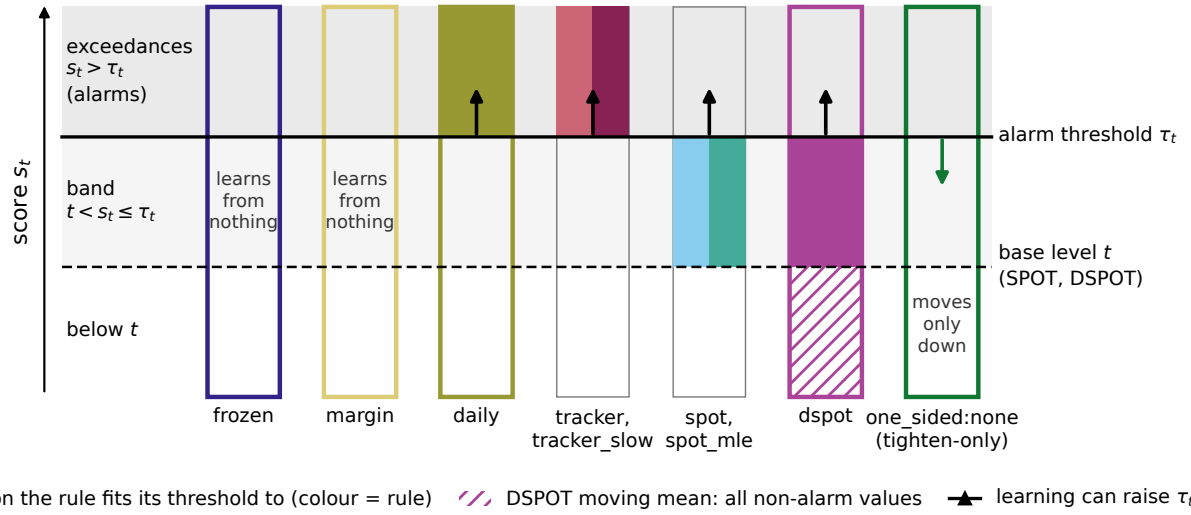


Figure 1: Learning regions of the threshold rules (schematic). The alarm threshold τ_t and, for SPOT and DSPOT, the base level t below it split the score range into exceedances ($s_t > \tau_t$), the band ($t < s_t \le \tau_t$) and values below t. A filled column segment marks the region a rule fits its threshold to; DSPOT's moving mean also absorbs every non-alarm value (hatched). Daily recalibration and the trackers learn from exceedances, so the attacks they are meant to detect raise their thresholds; frozen and the static margin learn from nothing; one_sided:none can only move downward. Data status: schematic with no data, so the registered / exploratory / post-hoc label does not apply; the rule behaviour follows Table 1.

blocks whenever the review window holds a labelled event, the rule caught the terminal step in 1 of 10 control placements on HAI 20.07, against 10 of 10 for frozen.

4 Monitor, rules and data

Monitor. D1 is the world-model residual monitor of earlier registered studies on HAI [25]: an action-conditioned ridge regression with LightGBM residual corrections (60 trees, history 3) predicts each sensor one and five steps ahead; the score is the mean of the three largest standardised absolute residuals at each horizon, summed and smoothed over 60 s. D0 is a model-free control: the mean of the three largest standardised first differences. Both are calibrated on the most recent 20% of normal training data: to 2 false-alarm episodes per day on the 1 Hz stream datasets (episodes merged across 60 s gaps), to a per-sample false-alarm target of 0.05 on voraus-AD, and to one false-alarm episode per week on BATADAL (episodes merged across 1 h). In tt1d one monitor is fitted per operating mode of Tennessee Eastman, calibrated on four of that mode's eleven development runs, and one for ProvICS; every duration is converted to the dataset's step; and a third score is a registered secondary: D2, the mean of the three largest standardised residuals after projection on the principal components that explain 95% of the variance.

Rules. frozen (τ_0 at the budget); margin (τ_0 + 0.25× the spread between the calibration 99.9th percentile and median); daily (trailing-day quantile at the calibration exceedance rate p); tracker

($\eta = 0.01\times$ spread on the stream datasets, $0.05\times$ spread on voraus-AD) and tracker_slow ($\eta/10$, the best false-alarm adherence among trackers in development); tighten-only with four automatic permits and an oracle permit that simulates an operator who refuses to loosen while an event is under way: eleven rules in all. On voraus-AD the spread is the calibration 99th percentile minus the median and daily recalibration happens every 50 samples. The streaming extreme-value rules (SPOT and DSPOT, with method-of-moments and maximum-likelihood tail fits [18, 31]) were not part of the tt1 registration: they enter in Section 7 as exploratory and post-hoc analyses and are registered in the BATADAL replication (tt1b) and, in their standard maximum-likelihood form, in tt1d.

Registration. The protocol fixed the monitor, splits, rules, injection designs, gates and seeds; the harness recorded one ledger entry per dataset before reading a test array, refused to run if a frozen file's SHA-256 differed from the registration, and computed the gates from its own output. A verifier checks that each access followed the registration commit, that the ledger holds no access the protocol does not allow, and that the stored gates match a recomputation (27 of 27 checks pass for tt1, 8 of 8 for tt1b, and 17 of 17 and 20 of 20 for the two datasets of tt1d). tt1d keeps two ledger entries per dataset, for its attack data and its sealed attack-free data; ProvICS has three of each: a failed first attempt, a count of missing values, and the amended attempt (Section 6.4).

Development. The registered claim of tt1 differs from the one we set out to test. We began with a tighten-only threshold that loosens through a statistical permit, moved to a static safety margin, and registered frozen against adaptive after the first two lost recall or missed the budget on the development releases (Appendix D).

5 Registered results

5.1 Primary: HAI 20.07

False-alarm episodes per day are counted over the 4.94 days of unlabelled time inside the 5.15-day test streams. G1 passes as registered, but on this stream the pulses were not needed: the adaptive rules also missed the terminal step in the control runs, because the test stream's real attacks had already raised their thresholds 7 to 26 times above τ_0. The pulse effect in isolation comes from development on attack-free normal data (frozen 30/30; daily 4/30, control 21/30; tracker 1/30, control 23/30; tracker_slow 3/30, control 25/30). Neither registered test of a deliberate campaign, G1 here or H3 on BATADAL (Section 6.1), could show an effect, because the rules under test were already blind in the controls; the campaign results in Section 7 are exploratory, and those of Section 6.3 registered.

5.2 Real attacks inflate adaptive thresholds

At HAI 20.07 attack onsets the median threshold stood at 25.6 τ_0 for daily recalibration, 12.9 for the tracker and 6.4 for the slow tracker. HyGrid's daily rule equals frozen because its 14.5 h final never completes a day.

On HAI 20.07 and BATADAL every disagreement between an adaptive rule and the frozen threshold went one way. On HAI 20.07 the frozen threshold caught 16 attacks the tracker missed and the tracker caught none that it missed (exact McNemar $p = 3.1\times10^{-5}$); the slow tracker split 9 to 0 ($p = 0.0039$) and daily recalibration 18 to 0 ($p = 7.6\times10^{-6}$). On BATADAL the tracker split 8 to 0 ($p = 0.0078$). These counts are post-hoc, from the recorded trajectories, which reproduce every registered event count.

Time in alarm and onset recall (post-hoc). Episode counts hide how long an alarm lasts. On HAI 20.07 the frozen threshold was in alarm for 39% of the unlabelled time. The D1 score sat above τ_0 for the first 23.8 h of test1, of which the first 2.7 h carry no attack label and seven attacks begin inside the rest, and for a 17-hour stretch of test2 that opens with an attack and holds two more. An alarm episode that overlaps an attack does not count as a false alarm, so the registered episode count records none of these hours. Ten of the frozen threshold's 36 detections were attacks that began while an alarm was already running, nine of them inside these two runs. Counting an attack only when an alarm episode starts inside it, the detection rule that the authors of the Sherlock dataset recommend [35], the frozen threshold detects 26 of 38, the tracker 20 (-0.16, $[-0.37, 0.05]$), the slow tracker 27 ($+0.03$, $[-0.18, 0.24]$) and daily recalibration 11 (-0.39, $[-0.55, -0.24]$). On BATADAL the frozen threshold was in alarm for 1.4% of the unlabelled time and the stricter count changes nothing (tracker -0.57, $[-0.79, -0.29]$). On HyGrid-ICS (4.1% in alarm) the 91 attacks cover 83% of the final rows, about 100 s apart on average, so one merged alarm episode spans several attacks and the stricter count credits few of them: frozen 3 of 91, tracker 8 ($+0.05$, $[0.00, 0.11]$), slow tracker 5. There the stricter count reverses the sign, but it measures how alarms merge across back-to-back attacks more than what the threshold detects. G2 stands as registered, but the tracker's HAI 20.07 gap depends on crediting the frozen threshold's long alarms; the BATADAL gap does not. An earlier post-hoc variant hid each file's opening run above τ_0 from the threshold update and gave tracker -0.29 $[-0.45, -0.16]$ and slow tracker -0.08 $[-0.18, 0.00]$; we superseded it because the opening run of test1 turned out to contain seven attacks.

5.3 False alarms

Where an adaptive rule raised fewer false alarms than frozen, it bought them with the recall losses of Section 5.2, except for the oracle permit on voraus-AD (0.064 against 0.081 at equal recall), which uses the attack labels. The one rule that cut false alarms at no recall cost here, the static margin on HyGrid-ICS (0.0 against 19.2 per day at identical recall), does not adapt. The margin matched frozen recall on HAI 20.07 and held voraus-AD's target at a cost of 12 points of recall there. Counted in time rather than episodes (post-hoc), the frozen threshold was in alarm for 39% of the unlabelled time on HAI 20.07, 4.1% on HyGrid-ICS and 1.4% on BATADAL; the tracker for 0.2%, 0.3% and 0.4%. Tighten-only equalled the frozen threshold on all three tt1 datasets (2.23, 19.2 and 0.081).

5.4 Other registered outcomes

The protocol commits us to report every rule, gate and failure; Appendix A gives the rest of the registered record. Every automatic loosening permit lost recall on HAI 20.07, and the oracle permit, which equalled frozen on the real attacks, caught the terminal step in 1 of 10 control placements of the G1 campaign.

Table 2: Datasets. Event counts are the labelled events of each final split.

Dataset	System	Training data	Final data	Events
HAI 20.07 (primary)	HIL testbed (boiler, turbine, water treatment)	547,624 s normal, 31 sensors, 20 actions	test1 + test2, 5.15 days	38 attacks
HyGrid-ICS	simulated hybrid grid SCADA, 5 Hz to 1 Hz	9.6 h (label-blind first 40%)	14.5 h	91 attacks
voraus-AD	industrial robot, 100 Hz to 10 Hz	948 normal samples	1174 samples	755 anomalous samples
HAI 21.03, 22.04, 23.05 (development)	HIL testbed	normal training data	spent test files	50, 58, 52 attacks
BATADAL (registered replication)	water distribution network, 15 min	384 days normal	two test files, 260 days	14 attacks
Tennessee Eastman (tt1d)	simulated chemical process, 7 modes, 3.6 s	11 attack-free runs of 120 h per mode	142 attack runs; 56 sealed attack-free runs	245 attacks
ProvICS (tt1d)	reactor twin under a real PLC, 2 s	39 h attack-free	21 h attack recording; 13 h sealed attack-free	32 events

Table 3: Registered gates of tt1 on HAI 20.07. Data status: registered.

Gate	Registered test	Result
G1	after a pulse campaign (12×120 s per day at twice the terminal magnitude), daily, tracker and tracker_slow detect the terminal step in at most (frozen $-$ 0.5) of placements; frozen at least 0.8	frozen 10/10, each adaptive rule 0/10: pass, but uninformative (the adaptive controls were already 0/10)
G2	recall(tracker) $-$ recall(frozen) on the real attacks, 95% paired event bootstrap upper bound below 0	0.526 vs 0.947, -0.42 $[-0.58, -0.26]$: pass
G3	frozen false-alarm episodes per day on unlabelled test time at most 4	2.23: pass (above the 2.0 budget; in alarm for 39% of the unlabelled time, post-hoc)

Table 4: Event recall and the paired difference against the frozen threshold (95% paired event bootstrap) on the D1 score. Data status: registered (tt1).

Dataset (events)	frozen	daily	tracker	tracker_slow	tighten-only
HAI 20.07 (38 attacks)	0.947	0.474 $[-0.63, -0.32]$	0.526 $[-0.58, -0.26]$	0.711 $[-0.37, -0.11]$	0.947
HyGrid-ICS (91 attacks)	0.297	0.297	0.154 $[-0.22, -0.08]$	0.165 $[-0.21, -0.07]$	0.297
voraus-AD (755 anomalous samples)	0.334	0.148 $[-0.21, -0.16]$	0.082 $[-0.28, -0.22]$	0.213 $[-0.14, -0.10]$	0.334

Table 5: False alarms. Data status: registered (tt1).

Dataset	frozen	margin	daily	tracker	tracker_slow
HAI 20.07 (episodes/day)	2.23	2.23	0.20	1.01	2.23
HyGrid-ICS (episodes/day, 2.5 h unlabelled)	19.2	0.0	19.2	19.2	19.2
voraus-AD (per-sample FPR, target 0.05)	0.081	0.033	0.086	0.012	0.014

6 Registered replications

6.1 BATADAL (tt1b)

We registered a second study (tt1b) to test the exploratory findings of Sections 7.1 and 7.5 on data no analysis here had opened: the C-Town network of BATADAL [33], whose original attacks conceal themselves in the SCADA readings, in the re-simulation without that concealment published with the artifact of Erba et al. [12], put on a 15-minute grid. All time-scale settings were fixed in physical time before access: 1 h smoothing, episodes merged across 1 h, one false-alarm episode per week, a one-day DSPOT mean. The test files hold 14 attacks in 260 days.

We retain the primary rule as registered and report H3 as uninformative rather than refuted. H2 passes under the registered bootstrap, whose interval is $[0, 0]$ because no event was discordant; an exact one-sided 97.5% bound for 0 discordant events in 14 is -0.23,

Table 6: Registered hypotheses of tt1b on BATADAL. Data status: registered.

Hypothesis	Registered test	Result
H1	recall(tracker) − recall(frozen), interval upper bound below 0	6/14 vs 14/14, −0.57 [−0.79, −0.29]: pass
H2	recall(MLE SPOT) − recall(frozen), lower bound at least $-\max(0.10, 2/N)$	14/14, difference 0: pass
H3	24 h closed-loop frog: DSPOT at least 0.5 below its control; frozen and tighten-only within 0.1	DSPOT already 0/10 in the control: fail
Primary rule	H1 and H2 and H3	fail

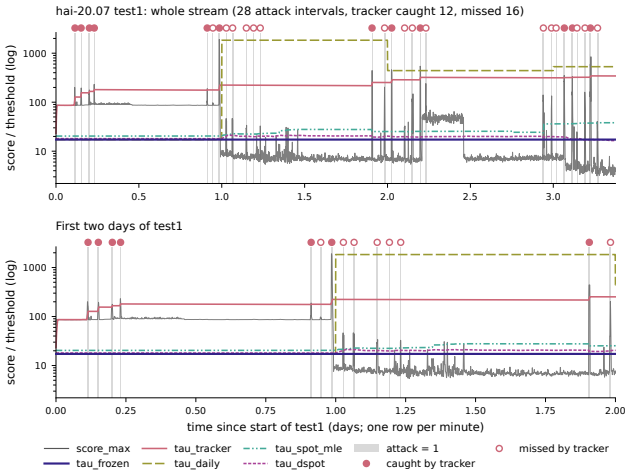


Figure 2: Threshold trajectories on the HAI 20.07 test1 stream (4,860 minutes). Grey: the D1 score score_max. Coloured: the frozen, tracker, daily, spot_mle and dspot thresholds, log scale. Grey shading marks attack-labelled minutes; each marker above the panel is one attack interval, filled if the tracker caught it. test1 holds 28 of the 38 attacks; the tracker caught 12. The score stays above $\tau_0 = 17.26$ for the first 23.8 h of test1, so the frozen threshold is in alarm throughout; only the first 2.7 h carry no attack label, and seven attacks begin inside the run. That unlabelled opening lifted the tracker to $5.0\,\tau_0$ before the first attack ($5.25\,\tau_0$ at its onset), and the early attacks then raised the tracker and daily thresholds far above frozen. Later attacks arrive under the inflated thresholds. Data status: post-hoc analysis; the trajectories were computed after the registered finals. The recomputed calibration threshold equals the registered one exactly. Over test1 and test2 the traces give frozen 36/38 and tracker 20/38 (registered tt1) and spot_mle 35/38 and dspot 36/38 (post-hoc: SPOT and DSPOT were not in the tt1 registration). test2 is not drawn; test1 has no missing values.

beyond the −0.14 tolerance, so H2 is not informative under an exact method. SWaT and WADI are registered under the same protocol. We requested them from their publisher on 2026-10-23 and had not been granted access when we submitted, so the registered replication rests on BATADAL alone; if access arrives, they will run once under the registration and be reported whatever they show.

Table 7: BATADAL, D1 score. Data status: registered (tt1b).

Rule (D1)	Recall	False alarms (× budget)	Onset inflation
frozen	14/14	0.70	1.0
tighten-only	14/14	2.43	1.0
SPOT (MLE)	14/14	0.81	1.0
daily	14/14	6.34	0.69
tracker	6/14	1.83	409
tracker_slow	5/14	0.70	74
SPOT (moments)	6/14	0.13	26
DSPOT	6/14	0.13	24
margin	3/14	0.05	173

H3 could not be tested because DSPOT's control was already blind, and the blindness predates the attacks. Fitted by the method of moments to BATADAL's heavy-tailed calibration scores (spread 6,462 against $\tau_0 = 9.39$), SPOT's alarm level started at $49.0\,\tau_0$ and DSPOT's at $45.8\,\tau_0$ at the first step of each test file, and both fell through the streams, to 24 to 44 τ_0 at the first attack (post-hoc trajectories; the registered per-event onset inflation agrees). Rerun with no attack sample in their update, SPOT still caught 6 of 14 and DSPOT 7 of 14 (post-hoc). MLE SPOT started at $1.22\,\tau_0$ and caught all 14. On BATADAL the moments-based extreme-value rules started too high because of their calibration fit; the attacks did not raise them.

The registered frog variants (10 placements each) did test MLE SPOT. Under the 24 h closed-loop frog it caught the terminal attack in 7 of 10 placements against 9 of 10 in its control with D1, and in 0 of 10 against 4 of 10 with D0; with 6 h and 1 h campaigns it matched its D1 control and caught 2 and 4 of 10 against 4 with D0. Frozen and tighten-only caught 10 of 10 in every variant, daily recalibration 9 of 10, the margin none, and the tracker, SPOT and DSPOT at most 1 of 10 with or without a campaign. The frozen threshold detected the 14 real attacks a median 3.5 steps (52.5 minutes) after onset and at most 65 steps (16.3 h) after.

With the model-free D0 score, MLE SPOT also lost recall (9/14, [−0.64, −0.14]). The margin and the tracker step are scaled by the calibration spread (99.9th percentile minus median); BATADAL's residual scores are heavy-tailed, so the margin sat 173 times above τ_0. Parameters scaled by a tail spread are fragile when the tail is heavy.

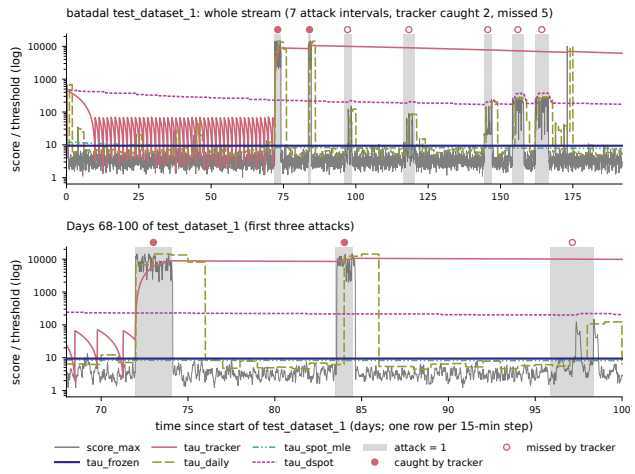


Figure 3: Threshold trajectories on the BATADAL test_dataset_1 stream (18,433 steps of 15 minutes, 192 days); layout as in Figure 2. test_dataset_1 holds 7 of the 14 attacks; the tracker caught 2. The lower panel zooms to days 68 to 100, which cover the first three attacks, because the first attack starts on day 72. score_max is missing for the first 10 steps ($t = 0$ to 9, 2.5 h); those steps are left as a gap and the thresholds are drawn there as given. The attacks lift the tracker to about 10^4. DSPOT's moments fit starts $45.8\,\tau_0$ above τ_0 (9.39) at $t = 0$, before any attack, and falls through the stream, while frozen and spot_mle stay near τ_0. Data status: post-hoc analysis of the tt1b dataset, which was opened once for the registered study; the recomputed τ_0 equals the registered value exactly. Over both test files the traces give frozen 14/14, tracker 6/14, spot_mle 14/14 and dspot 6/14, the registered tt1b counts. test_dataset_2 is not drawn.

6.2 Tennessee Eastman: attacks that follow attacks (tt1d)

A third study registered what the first two left open: both detection counts, the mechanism, the stronger rules and ceilings of Sections 7.6 and 7.7, and campaigns whose controls can pass or fail. Its main dataset is Kaspersky's Tennessee Eastman attack set [10, 13]: 142 attack runs in seven operating modes and attack-free runs of 120 h, one row every 3.6 s. File names fixed each run's role before any content was read. Per mode, eleven attack-free runs fit and calibrate one monitor and eight more stayed sealed. Two attack types hold one attack per run and two hold several in a row: three and four in the dataset's description, two or three events here, because back-to-back attacks with the same indicators merge under our event definition. An event is either the *first* of its run (141 events) or a *follower* (104), and every follower begins within one step of the end of the attack before it.

H1a passed. The tracker caught 66 of the 104 followers against 104 for the frozen threshold (-0.37, $[-0.47, -0.27]$, whole runs resampled; all 38 discordant events favour frozen), and daily recalibration 59 ($[-0.55, -0.32]$). *H1c passed*: on first events the tracker caught 116 against 116 ($[-0.05, +0.05]$; six discordant events each way). The tracker's loss falls on attacks that follow attacks and not on the first attack of a run, which is the mechanism of Section 7, here registered, and it sits in the runs of consecutive noise attacks (57 of 95 events against 95). Daily recalibration also lost first events (106 against 116, $[-0.12, -0.02]$). *H1b failed*: under the onset count the tracker caught 37 followers against 44 (-0.07, $[-0.13, 0.00]$, the upper bound exactly zero), so the registered primary rule, H1a and H1b, failed. The registration had named the reason in advance: a follower begins while the alarm of the attack before it is still running, and the onset count then credits the frozen threshold with only 44 of the 104 followers it alarmed on. *H2 passed*: the excluding recalibration and the capped tracker each caught all 104 followers. H3, the ceiling's consistency with Proposition 4, held on every event. With D0 (secondary) the tracker again lost followers and not first events ($[-0.30, -0.10]$ and $[-0.06, +0.07]$). With D2 it tightened instead: on the unlabelled time of the attack runs it ran at 33 times the false-alarm budget, against 1.7 for frozen, and caught 134 first events against 79, with a follower difference of $[-0.11, +0.03]$. Appendix E lists every registered rule.

Table 8: Tennessee Eastman, D1: events detected, and false-alarm episodes over the budget on the 56 sealed attack-free runs (280 days). Data status: registered (tt1d).

Rule	First (141)	Followers (104) any alarm	Followers (104) onset	False alarms
frozen	116	104	44	0.98
tracker	116	66	37	1.21
tracker, $\eta = 0.01\,\tau_0$	123	55	37	1.64
tracker_slow	110	88	50	0.83
daily	106	59	40	1.22
daily, episodes left out	113	104	44	1.33
tracker, capped rise	130	104	44	2.12
tracker, ceiling $1.25\,\tau_0$	121	104	41	1.30
SPOT (ML)	106	104	43	0.84
tighten-only	122	104	44	1.57

6.3 Tennessee Eastman: registered campaigns

Each of the 56 sealed attack-free runs received one campaign placement, drawn after its first 12 h: a 24 h campaign at the score level, one quiet hour, then a 600 s terminal bump (Table 9). The loud campaign sets the score to $2\,\tau_0$ for 120 s every 2 h; the frog and the sub-event pulses are those of Section 7.7, with every duration rounded to the 3.6 s step (a sub-event pulse lasts 43 s).

All three campaign hypotheses passed. *H4*: after the frog the standard SPOT caught 16 of 56 bumps of $1.25\,\tau_0$ against 56 in its control ($[-0.82, -0.59]$), and with a ceiling at $1.25\,\tau_0$ all 56 bumps of $1.5\,\tau_0$. In the rehearsal SPOT's own level started up to $1.06\,\tau_0$, which is why H4 was registered at $1.25\,\tau_0$ and not $1.05\,\tau_0$. *H5*: after the loud pulses the tracker caught none of 56 bumps of $1.05\,\tau_0$ against 45 in its control, and the excluding recalibration was unchanged (49 against 49). *H6*: the sub-event pulses left the excluding recalibration and the hold-off tracker with none of 56, against 49 and 47, while both ceilings caught all 56 bumps of $1.5\,\tau_0$ (in 4 of the 56 placements an alarm episode of the excluding rule still

Table 9: Campaigns on the 56 sealed attack-free Tennessee Eastman runs: terminal bumps of 1.05 / 1.25 / 1.5 τ_0 caught after the campaign, of 56. Without a campaign the adaptive rules caught 44 to 51 bumps of $1.05\,\tau_0$ and 55 or 56 of the larger ones; the frozen threshold caught all 56. Data status: registered (tt1d).

Rule	24 h frog	Loud pulses	Sub-event pulses
frozen	56 / 56 / 56	56 / 56 / 56	56 / 56 / 56
tracker (as registered in tt1)	–	0 / 0 / 0	0 / 0 / 0
daily recalibration	–	0 / 0 / 0	0 / 0 / 0
daily, alarm episodes left out	47 / 56 / 56	49 / 56 / 56	0 / 0 / 0
tracker, rise stopped after 60 s of alarm	–	0 / 0 / 0	0 / 0 / 0
tracker, rise capped at $0.1\,\tau_0$ per day	51 / 56 / 56	8 / 55 / 56	6 / 55 / 56
SPOT, maximum likelihood	0 / 16 / 24	44 / 56 / 56	44 / 56 / 56
SPOT, ceiling $1.25\,\tau_0$	0 / 16 / 56	–	–
tracker, ceiling $1.25\,\tau_0$	46 / 56 / 56	0 / 56 / 56	0 / 56 / 56
daily, ceiling $1.25\,\tau_0$	–	0 / 0 / 56	0 / 0 / 56
SPOT, approval every 6 h / daily	1 / 17 / 24, 32 / 42 / 53	–	–

reached event length). The frozen threshold caught every bump in every campaign and control. The quiet-log approval gate (secondary) caught 1 of 56 bumps of $1.05\,\tau_0$ after the frog with a review every 6 h and 32 with a daily review, against 44 in its control. Over the 280 sealed attack-free days the frozen threshold ran at 0.98 times its false-alarm budget (Table 8); the adaptive rules ran above it, from 1.21 for the tracker to 2.12 for the capped tracker, except SPOT (0.84).

6.4 ProvICS (tt1d)

The second dataset of tt1d is ProvICS [28]: a chemical reactor twin under a real PLC, with 32 labelled events in a 21 h recording. The last quarter of its attack-free recording was sealed. The registered run stopped with an error before any result was printed or written, because one sealed row lacks its setpoint values; a ledgered read then counted missing values and nothing else, and a dated amendment, committed before the second attempt, repeats the previous value in an empty cell and changes nothing else.

P1 failed: the tracker detected exactly the events the frozen threshold detected, 25 of 32 under the any-alarm count and 4 under the onset count. The monitor did not hold its calibration on this recording. The frozen threshold was in alarm for 44% of the unlabelled time, at 33 times its budget, so most of its detections are events inside standing alarms; the tracker reached the same recall in alarm for 19% of the unlabelled time, at 6.5 times the budget. Here adaptation cost no recall and removed false alarms. P2 and P3 held: the capped tracker and every ceiling rule detected the same 25 events. With D2 (secondary) the tracker lost 9 of the 25 events ($[-0.44, -0.12]$). On the sealed rows (six overlapping placements of a 6 h campaign, descriptive) the loud pulses left the tracker with none of 6 bumps of $1.05\,\tau_0$ and the sub-event pulses left the hold-off tracker with none of 6, but the frog did not move SPOT, whose level rose by 0.6% and which caught all 6 (P4 failed).

7 Learning regions under attack (exploratory)

The analyses in this section were not registered. They use the spent HAI releases for development and reopen the three tt1 datasets after their finals; every access is in the ledger.

7.1 Extreme-value thresholds escape natural poisoning on the HAI data

On the three spent releases the moments SPOT, the maximum-likelihood SPOT and DSPOT stayed within two events of the frozen threshold on each release (recall 0.58 to 1.00 against 0.60 to 1.00), where the tracker lost 22 to 37 points; post-hoc on HAI 20.07 they caught 36, 35 and 36 of 38 attacks against 36 (Table 13 in Appendix C).

The exceptions: on HyGrid-ICS the moments SPOT lost 9 points ($[-0.15, -0.03]$); on voraus-AD, 190 calibration samples leave about four tail points above SPOT's base level (the calibration 98th percentile), and SPOT ran at 32% false positives against a 5% target. On the HAI data the extreme-value thresholds started near calibration: at the start of both HAI 20.07 test files the moments SPOT stood at $1.09\,\tau_0$, DSPOT at 1.03 to 1.05 and MLE SPOT at 1.18 (post-hoc). Section 6.1 shows what a heavier calibration tail does to that starting point.

7.2 The effect does not depend on the score

With D0 the tracker lost recall on every dataset where it was measured: $[-0.38, -0.14]$, $[-0.21, -0.05]$ and $[-0.19, -0.02]$ on the spent releases (exploratory), $[-0.32, -0.08]$ on HAI 20.07, $[-0.12, -0.02]$ on HyGrid-ICS and $[-0.04, -0.01]$ on voraus-AD (post-hoc), and $[-0.86, -0.36]$ on BATADAL (registered tt1b). With the reconstruction score D2 (exploratory here), the frozen threshold caught 65 of the 160 events of the development releases and the tracker 53 ($[-0.14, -0.01]$) pooled; on HAI 22.04 the tracker tightened instead and caught 18 against 11, at 31 false alarms a day.

7.3 Mechanism: attacks that follow attacks

The tracker's misses concentrate on attacks that start within 6 h of the previous attack's end. D1 recall on close versus isolated attacks was 0.75 / 1.00, 0.41 / 1.00 and 0.20 / 0.67 on the spent releases and 0.48 / 0.71 on HAI 20.07 (31 close, 7 isolated). The frozen threshold also detected close attacks less often on two releases, so part of

the gap reflects harder attacks; the tracker's gap is larger on every release.

Attacks are the main thing that raises the tracker. Rerun post-hoc with its update fed only the attack-labelled samples (the others replaced by the calibration median for the update, alarms still on the real score), the tracker detected 20 of 38 on HAI 20.07 and 14 of 91 on HyGrid-ICS, exactly its registered counts, and 7 of 14 on BATADAL (6 registered). Fed only the unlabelled samples, it detected 28 of 38, 27 of 91 (equal to frozen) and 11 of 14: the unlabelled exceedances add a smaller loss of their own, -0.21 $[-0.34, -0.08]$ on HAI 20.07 and -0.21 $[-0.43, 0.00]$ on BATADAL.

7.4 Loud poisoning: pulse campaigns

Normal data of the spent releases, 30 placements; the terminal step is the smallest setpoint step the frozen threshold detects on the clean stream. After a campaign of 120 s supra-threshold pulses, twelve a day for up to a day, daily recalibration caught 4 of 30 terminal steps (control 21), the tracker 1 (23) and the slow tracker 3 (25); the frozen threshold caught all 30 and SPOT 29 (Table 14 and Figure 4 in Appendix C).

Open-loop sub-alarm pulses did not poison SPOT: its threshold fell (0.82 to 0.98 τ_0), and many pulses crossed SPOT's own alarm level and were refused.

7.5 Silent poisoning: the closed-loop frog

Score-level attacker A2, normal data of the spent releases, 10 placements each: the attacker holds the score at $0.97\,\tau_t$, then, after a one-hour lead, a 600 s bump at $1.05\,\tau_0$ follows (600 s is the terminal length of every development campaign). After 24 h the moments SPOT and DSPOT caught none of 30 terminal bumps and the maximum-likelihood SPOT 8; the frozen threshold, the slow tracker and tighten-only were unaffected (Table 15).

Each rule is compared with its own control (the same bump without a campaign). The frozen and both SPOT controls caught 30/30 and DSPOT's 24/30; the tracker and daily controls already missed some bumps on the drifting releases, and the campaign did not lower them further. After 1 h the moments SPOT threshold stood at 1.1 to 4.5 τ_0; after 24 h it had grown by more than five orders of magnitude, and the MLE fit left the floating-point range on one release. A synthetic control with a light tail (the 60-step moving average of $|N(0,1)|$, five seeds) did not run away: the moments SPOT threshold stayed at 0.95 to 0.97 τ_0 and caught every terminal bump, and maximum-likelihood SPOT rose to a median of 1.02 to 1.04 τ_0 (at most 1.06), though it missed the bump on one of the five seeds in the 24 h, 6 h and fixed-level variants. The runaway therefore depends on the heavy tail of real residual scores.

A physically realizable version. We also gave the attacker only the readings of three sensors coupled to one setpoint channel, spoofed with zero-mean noise that keeps the D1 score at 0.97 of the rule's threshold for 6 h (Appendix B). The maximum-likelihood SPOT caught 4 of 12 terminal steps against 12 in its control and the moments version 6; DSPOT was unchanged (9 against 9). On HAI 22.04 the median campaign raised no alarm; on HAI 23.05 it was not silent.

7.6 Stronger baselines and a standard SPOT (exploratory)

The rules above are the plain versions. After the mock review we fixed four stronger adaptive rules and three SPOT variants in a written plan and ran them on the spent development releases (the plan was committed before the runs; one rule was corrected by a dated amendment after the first run, and every run was then repeated). They are daily recalibration that leaves out alarm episodes longer than 60 s and the hour after them; the tracker with its upward steps stopped once an alarm has lasted 60 s and for an hour after; a tracker whose rise is capped at $0.1\,\tau_0$ per day; Telemanom's dynamic threshold [19]; SPOT with a maximum-likelihood tail refitted as the peaks grow by 1%, also with the shape bounded to $|\xi| \le 0.5$ and with runs declustering; and DSPOT on the maximum-likelihood tail (Table 16 in Appendix C).

Leaving events out of the update, or capping the rise, removed natural poisoning on these releases: the excluding rule matched frozen recall and the capped tracker missed one more event. Stopping the tracker after a minute of alarm did not, because by Proposition 1 the first minute of each attack still keeps it raised for about a day. The excluding rule also resisted the pulse campaign, whose 120 s pulses count as events. An attacker who knows the 60 s event length can keep each exceedance shorter; Section 7.7 runs that attack. The capped tracker lost a little to the pulses, and its cap bounds how fast it rises, not how far. Telemanom's threshold caught more attacks at 16 to 19 times the false-alarm budget, so its recall is not comparable with the others.

The standard SPOT did not escape the closed-loop attacker. Refitted by maximum likelihood, with or without a bounded shape, it missed the terminal step in all 30 placements of the 24 h campaign, and DSPOT on the same tail in all 30. Declustering resisted a continuous frog, which forms a single cluster, but an attacker who leaves the score clean for two minutes after each minute of pushing blinded it on HAI 22.04 (10 of 30 pooled, against a control of 20).

7.7 Bounded loosening, sub-event pulses and an approval gate (exploratory)

A second written plan, committed before its runs, tested three things on the same releases. *A ceiling*: the tracker, daily recalibration and the standard SPOT with the deployed threshold clipped at $\kappa\tau_0$ (Proposition 4). *Sub-event pulses*: a score-level attacker who knows the 60 s event length sets the score to twice the current threshold for 45 s every 30 minutes for a day, so that a pulse by itself is too short to count as an event. *An approval gate*: SPOT proposes its level, and a ceiling rises to it at each review if the review window held no alarm, a stand-in for an operator who signs a loosening when the alarm log is quiet. Campaigns end in terminal bumps of 1.05 to $10\,\tau_0$ (Table 10).

With a ceiling at $1.1\,\tau_0$ the tracker and daily recalibration caught the 125 events the frozen threshold caught. Wider ceilings cost recall, as Proposition 4 allows: at 1.25 and $1.5\,\tau_0$ the tracker caught 118 and 108, against 116 and 107 for the static thresholds at those levels. The frog shows what a ceiling buys. Without one, the standard SPOT missed every terminal bump up to $1.5\,\tau_0$, and on two of the three releases every bump up to $10\,\tau_0$, because its threshold

Table 10: Ceilings, sub-event pulses and an approval gate on the three spent HAI releases. Natural poisoning: events detected of 160. Campaigns: terminal bumps of 1.05 / 1.5 / 3 τ_0 caught after the campaign, of 30 placements; where a rule's control at 1.05 τ_0 was below 30 it is given in parentheses. Data status: exploratory.

Rule	Natural poisoning	24 h frog	Sub-event pulses
frozen	125	30 / 30 / 30	30 / 30 / 30
tracker (as registered)	77	–	0 / 0 / 0 (20)
tracker, ceiling 1.1 / 1.25 / 1.5 τ_0	125 / 118 / 108	–	6 / 30 / 30 (25), 0 / 30 / 30 (20), 0 / 30 / 30 (20)
daily recalibration	104	–	0 / 5 / 24 (18)
daily, ceiling 1.1 / 1.25 / 1.5 τ_0	125 / 121 / 119	–	0 / 30 / 30 (18), 0 / 30 / 30 (18), 0 / 5 / 30 (18)
daily, alarm episodes left out	125	30 / 30 / 30	0 / 11 / 30
tracker, rise stopped after 60 s of alarm	89	as control	0 / 0 / 0 (21)
tracker, rise capped at 0.1 τ_0 per day	124	30 / 30 / 30	13 / 30 / 30
SPOT, maximum likelihood	126	0 / 0 / 10	30 / 30 / 30
SPOT, ceiling 1.1 / 1.25 / 1.5 τ_0	126 / 126 / 126	0 / 30 / 30, 0 / 30 / 30, 0 / 7 / 30	–
SPOT, approval every 6 h / daily	127 / 127	15 / 19 / 30, 25 / 30 / 30	–

left the floating-point range. With a ceiling it missed only bumps at or below the ceiling and caught all 30 above it.

The sub-event pulses defeated both rules that depend on recognising an event. The excluding recalibration caught none of 30 terminal bumps up to 1.25 τ_0 (control 30) and the tracker with a hold-off none at any size up to 10 τ_0, since the pulses were too short to be left out. The capped tracker missed 17 of 30 at 1.05 τ_0 and none from 1.25 τ_0. SPOT, which does not learn from alarms, was unaffected. The quiet-log gate did not stop the frog: with a review every 6 h it caught 15 of 30 bumps at 1.05 τ_0 and 20 of 30 at 2 τ_0; with a daily review, 25 of 30 at 1.05 τ_0 and all 30 above. A campaign that raises no alarm passes an approval that reads the alarm log.

8 Design guidance

A threshold rule's learning region is its attack surface. Every automatic loosening path of the plain rules could be pushed upward: exceedance tracking and daily recalibration by real attacks (registered) and by loud pulses (exploratory); tail refitting and moving-mean drift tracking by a closed-loop attacker (exploratory); the spread and shift permits by pulses and benign drift in development, and on the registered HAI 20.07 stream all four lost recall (Section 5.4). The rules that held were the ones that never loosen themselves: frozen, and tighten-only without a permit. An operator-gated permit is only as good as the operator; our stand-in, which blocked loosening only while a labelled event was in its review window, caught 1 of 10 terminal steps in the G1 controls. A static margin never loosens either, but it is not safe by construction: on BATADAL its spread-scaled offset sat 173 times above τ_0 and it caught 3 of 14 attacks, as large a recall loss as any we measured.

Two stronger rules that do loosen themselves, daily recalibration that leaves out alarm episodes and a tracker with a capped daily rise, came within one event of frozen recall on the development releases (Section 7.6). Each still loosens through what it does not exclude: pulses shorter than its event length blinded the first, and the second rises by as much as its cap allows each day (Section 7.7). Telemanom's threshold avoided the losses only at 16 to 19 times the false-alarm budget.

Parts of this advice are established. Process-alarm standards already route alarm changes through authorised management of change [11, 20], lowering the threshold has been proposed against poisoning of retrained ICS detectors [23], and a bounded false-positive rate limits what a poisoner can gain [22]. What we add is the asymmetry for adaptive thresholds, automatic tightening with gated loosening, and its evaluation against natural, loud and silent poisoning.

We therefore recommend:

(1) **Let thresholds tighten automatically, never loosen.** A tracker capped at the calibrated threshold cannot lose an event the frozen threshold catches (Proposition 3) and can only add sensitivity in quiet periods.
(2) **If a threshold must loosen by itself, give it a ceiling.** Under a ceiling $\kappa\tau_0$ no poisoning can hide an event above $\kappa\tau_0$ (Proposition 4). On the development releases a ceiling of 1.1 τ_0 cost no recall on the recorded attacks, and in every campaign a ceiling hid only bumps at or below it.
(3) **Route loosening beyond the ceiling through an operator** who confirms from the process, not from the alarm log, that the plant is in normal operation: a gate that approved whenever the log was quiet let the silent campaign through. Treat a request to loosen during or soon after an alarm as suspect.
(4) **Recalibrate offline on verified clean data** [8], as a change-controlled event, when drift raises false alarms. Our frozen thresholds held 0.66 to 1.48 episodes per day on the three development test periods and 2.23 on HAI 20.07 against a budget of 2, but on HAI 20.07 the threshold was in alarm for 39% of the unlabelled time, the kind of drift an offline recalibration should catch.

(5) **Watch the threshold.** Log τ_t; a threshold that rises during an incident, or climbs steadily without alarms, is itself an indicator.
(6) **Budget alarm time as well as episodes.** On HAI 20.07 a 24-hour alarm that overlapped seven attacks did not count against the episode budget at all, and an attack that begins inside a running alarm gives the operator nothing new to see.

The cost is false alarms. On BATADAL the tighten-only rule ran at 2.43 times the budget against 0.70 for the frozen threshold, and over 280 attack-free days of Tennessee Eastman at 1.57 against 0.98 (both registered); on a development split of HAI 23.05 at 5.98 against 4.05, where a ten times slower tracker under the cap brought it to 4.05 (exploratory). A frozen threshold is not always the cheaper one: on ProvICS it ran at 33 times its budget where the tracker ran at 6.5 with the same recall. A static margin can cut that cost: on the HAI 23.05 development split the margin of our development rounds (half the calibration exceedance rate) ran at 0.58 times the budget. The registered margin (τ_0 + 0.25 times the spread) cost recall on voraus-AD (12 points) and BATADAL (11 of 14 attacks); a margin should not be scaled by a calibration spread that a heavy tail sets.

9 Limitations

- Three registered primary rules failed: tt1b's through H3, and both of tt1d's, through the onset count on Tennessee Eastman and through P1 on ProvICS. The registered support is the individual hypotheses that passed. The tracker's loss survives the onset count on BATADAL only (14 attacks), and that count is post-hoc there and on HAI 20.07.
- HAI 20.07 shares a testbed with the development releases. Tennessee Eastman is a simulated process whose attacks are large manipulations, and the registered campaigns are confirmed there only; on ProvICS the frog did not move SPOT, and the campaigns on HAI are exploratory.
- The stronger rules, the standard SPOT, the ceilings and the sub-event pulses were developed on HAI and registered on Tennessee Eastman; the registered rules of tt1 are the plain versions.
- The ProvICS result is registered with one dated amendment after a failed first attempt (Section 6.4).
- On HAI 20.07 the monitor itself drifted: the frozen threshold was in alarm for 39% of the unlabelled time, which inflates its recall under the any-alarm count (Section 5.2).
- HyGrid-ICS is a single simulated day with 2.5 h of unlabelled final time; voraus-AD anomalies are not attacks. Post-hoc, 0.15% of HyGrid-ICS's label-blind development rows (251 short runs) carry an attack label, against 83% of its final rows, so its calibration is very slightly contaminated.
- The pulse campaigns and both frogs are synthetic. The score-level frog sets the score directly and reads τ_t; the physically realizable version also runs the monitor itself on the spoofed history to choose each noise scale. Both are white-box upper bounds.
- SPOT's behaviour depends on the tail estimator and on the score's tail shape; we report both.
- Parameters scaled by a calibration spread (the static margin and the tracker step) are fragile on heavy-tailed scores: on BATADAL the margin sat 173 times above τ_0 and missed 11 of 14 attacks.
- Event recall counts any alarm inside a labelled interval, so it credits an attack that falls inside an alarm already running; on HAI 20.07 that gave the frozen threshold 10 of its 36 detections (Section 5.2). We do not model operator response time.
- Our monitor is one world-model residual, one model-free score and a linear reconstruction score (exploratory on HAI, a registered secondary in tt1d); nonlinear autoencoder detectors were not tested.
- We did not test a human operator; the approval gate of Section 7.7 is an automated stand-in. Human-approved suppression of detection rules has been observed to accumulate and persist [9], so routing loosening through people does not by itself stop a threshold from ratcheting upward.
- No claim of runtime authority, deployment safety or incident prevention follows from these results.

10 Conclusion

On HAI 20.07, BATADAL and Tennessee Eastman (registered), a quantile tracker that learns from exceedances missed attacks that a frozen threshold caught, and on Tennessee Eastman its misses fell on attacks that follow attacks and not on the first attack of a run. On BATADAL the loss survives counting only alarms that start inside an attack; on HAI 20.07 and Tennessee Eastman it does not, because later attacks arrive inside alarms that are already running, and on ProvICS, where the monitor was in alarm for 44% of the unlabelled time, the tracker lost nothing. How long a tracker stays raised follows from its calibrated false-alarm rate alone, so making it slower does not make it forget faster.

Deliberate poisoning was confirmed on sealed attack-free runs: loud pulses blinded the tracker, sub-event pulses blinded the recalibration that leaves alarms out, and a closed-loop attacker blinded a standard SPOT, while the frozen threshold caught every later attack and held its false-alarm budget over 280 days. On Tennessee Eastman every rule we tested that loosens itself was pushed through the part of its input it learns from; on ProvICS the frog did not move SPOT. A threshold that only tightens itself keeps the frozen threshold's recall by construction, and a ceiling confines what any of these attacks can hide to events at or below it. The price is false alarms: on Tennessee Eastman tighten-only ran at 1.57 times the budget and the tracker under a ceiling of 1.25 τ_0 at 1.30, against 0.98 for frozen.

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Ethical considerations

All experiments ran offline on public recordings: HAI 20.07 to 23.05, HyGrid-ICS, voraus-AD, the BATADAL re-simulation, the Tennessee Eastman attack set and ProvICS, each used under its publisher's terms. No live plant, testbed, vendor product or operator was involved, and no human subjects took part. The pulse campaigns and both frog attacks were injected into recorded normal data or into the score; the physically realizable frog changes recorded sensor values in software and is scored by our own monitor.

The attacks are not new in kind. Boiling-frog poisoning of self-updating detectors has been public since 2009 [6, 26], and the threshold rules we attack are textbook algorithms. What the paper

Table 11: Tighten-only with a loosening permit on HAI 20.07. Data status: registered (tt1).

Permit	Recall	vs frozen (95%)	FA/day	Permits/wk
spread, daily review	26/38	$[-0.39, -0.13]$	1.62	2.7
spread, 6 h review	28/38	$[-0.34, -0.08]$	2.63	4.1
shift, daily review	32/38	$[-0.21, -0.03]$	1.01	4.1
shift, 6 h review	31/38	$[-0.24, -0.03]$	2.02	12.2
oracle	36/38	$[0, 0]$	2.23	0

adds for an attacker is the observation that an adaptive threshold widens the window for a later attack; what it adds for a defender is a rule that closes that window at a known false-alarm cost. We judged that publishing both helps defenders more than attackers, because the defence is a configuration change (cap the threshold at its calibrated value) that any operator of an adaptive threshold can apply today. The released code generates attacks only against recorded data and a score file; it contains no tooling for reaching a control network.

Open science

The three pre-registrations, the data-access ledger, the attack and threshold harnesses, the registered-gate verifiers and the figure data and plotting code are available in the public repository at https://github.com/VyomKulshrestha/Ferrum-OS (paper sources, results and verifier reports under `docs/research`), which also contains the result files behind every number in this paper and the verifier reports (27 of 27 checks for tt1, 8 of 8 for tt1b, 17 of 17 and 20 of 20 for tt1d). The raw datasets are not redistributed here and are available from their publishers: HAI from its dataset repository, BATADAL from Taormina et al. [33] and the no-concealment re-simulation from the artifact of Erba et al. [12], voraus-AD from Brockmann et al. [4], HyGrid-ICS from its Zenodo record [37], the Tennessee Eastman attack set from Kaspersky [13] and ProvICS from its authors [28], under whose terms it is not redistributed. Reproducing the registered results requires one access to each raw dataset under the access rules recorded in the ledger. The repository's commit history shows that every registration was committed before the data access it governs.

A Other registered outcomes of tt1

Table 11 and the list below complete the registered record of tt1.

- **Automatic permits.** Every automatic loosening permit lost recall on HAI 20.07. On voraus-AD all four lost recall too (95% intervals: spread $[-0.21, -0.15]$ with daily review and $[-0.28, -0.21]$ with 6 h review; shift $[-0.18, -0.13]$ and $[-0.21, -0.15]$); on HyGrid-ICS only the 6 h shift permit did (24 of 91, $[-0.08, 0.00]$).
- **Oracle permit.** On the real attacks it equalled frozen, but in the G1 campaign runs it caught the terminal step in 7 of 10 placements after the pulses and 1 of 10 in the control.

Table 12: The physically realizable frog. Data status: exploratory (normal data of the spent releases).

Rule	Terminal steps caught, after campaign / in control (of 12 placements)	Threshold at the terminal step (median per release)
frozen	12 / 12	1.0 τ_0
SPOT (MLE)	4 / 12	1.04 to 1.59 τ_0
SPOT (moments)	6 / 12	0.78 to 1.42 τ_0
DSPOT	9 / 9	0.87 to 1.12 τ_0

- **First and second half of the HAI 20.07 events.** Recall was 0.947 / 0.947 for frozen, 0.474 / 0.579 for the tracker, 0.579 / 0.842 for the slow tracker and 0.474 / 0.474 for daily recalibration.
- **E5, sub-threshold noise campaign.** Only 1 of 13 placements admitted a stealthy noise level; 12 had none, so E5 gives no result.
- **BATADAL frog variants** are in Section 6.1. A separate registration that shared the HAI 20.07 and HyGrid-ICS access (detection-v2) is reported on its own.

B The physically realizable frog

The score-level attacker sets the score directly. We also gave the attacker only the readings of three sensors coupled to one setpoint channel. Every 5 minutes it reads the rule's threshold and spoofs the largest zero-mean noise whose D1 score, computed by the real world model on the spoofed history, stays at 0.97 of that threshold; after 6 h it stops, and 1 h later the just-detectable setpoint step follows. This attacker is white-box: it can run the monitor.

Controls caught 12 of 12 for frozen and both SPOT variants and 9 of 12 for DSPOT, so the campaign did not change DSPOT's total (per release, campaign against control: 4 against 3, 3 against 4 and 2 against 2). On HAI 22.04 both SPOT variants missed the terminal step on all four placements, and the median campaign raised no alarm. On HAI 23.05 it was not silent: the frozen threshold raised a median of 190 alarm steps during the campaign, and the SPOT variants 154 to 171. The spoofed noise had a scale of 0.02 to 0.04 robust units against frozen and the SPOT variants, and 0.05 to 0.12 against DSPOT, whose moving mean follows the noise-driven rise, so the attacker's cap rises with the mean rather than with the tail.

C Development experiments in detail

Tables 13 to 16 and Figure 4 give the per-rule results of the exploratory experiments of Section 7. Figure 5 collects the recall differences of tt1, tt1b and the development releases with their status.

Table 13: Event recall of the extreme-value rules. Data status: exploratory on the spent releases, post-hoc on HAI 20.07.

Data	frozen	tracker	SPOT (moments)	SPOT (MLE)	DSPOT
HAI 21.03 / 22.04 / 23.05	1.00 / 0.76 / 0.60	0.78 / 0.45 / 0.23	1.00 / 0.79 / 0.58	1.00 / 0.79 / 0.60	1.00 / 0.79 / 0.60
HAI 20.07 (post-hoc)	0.947	0.526	0.947	0.921 $[-0.08, 0.00]$	0.947

Table 14: Terminal attacks caught after a pulse campaign (control in parentheses). Data status: exploratory (normal data of the spent releases).

Campaign	frozen	daily	tracker	tracker_slow	SPOT	DSPOT
12×120 s supra-threshold pulses per day	30/30 (30/30)	4/30 (21/30)	1/30 (23/30)	3/30 (25/30)	29/30 (30/30)	23/30 (25/30)
24×300 s sub-alarm pulses per day, open loop	30/30	23/30 (23/30)	19/30 (21/30)	23/30	29/30	22/30 (24/30)

Table 15: Terminal attacks caught after a closed-loop frog campaign, 30 placements. Data status: exploratory (normal data of the spent releases).

Rule	24 h campaign	6 h	1 h	open loop, fixed level, 24 h
frozen, tracker_slow, tighten-only	as control	as control	as control	as control
tracker	as control	as control	as control	16/30 (control 20/30)
SPOT (moments)	0/30	0/30	0/30	not moved; alarms throughout
DSPOT	0/30	0/30	0/30	not moved; alarms throughout
SPOT (MLE, refit every 2048 peaks)	8/30	8/30	13/30	11/30

Table 16: Stronger adaptive rules and a standard SPOT on the three spent HAI releases. Natural poisoning: events detected of 160 and the paired difference against frozen (125 of 160). Pulses and frog: terminal steps caught after the campaign / in the control, of 30 placements. Data status: exploratory.

Rule	Natural poisoning	Pulses	24 h frog
tracker (as registered)	77, $[-0.37, -0.23]$	1 / 23	as control
daily recalibration	104, $[-0.19, -0.08]$	4 / 21	as control
daily, alarm episodes left out	125, $[-0.02, 0.02]$	30 / 30	30 / 30
tracker, rise stopped after 60 s of alarm	89, $[-0.29, -0.16]$	2 / 24	21 / 21
tracker, rise capped at $0.1\,\tau_0$ per day	124, $[-0.02, 0.00]$	25 / 30	30 / 30
Telemanom dynamic threshold (33 to 37 false alarms a day)	140, $[+0.04, +0.15]$	27 / 26	28 / 28
SPOT, maximum likelihood	126, $[-0.01, +0.03]$	30 / 30	0 / 30
SPOT, maximum likelihood, $\lvert\xi\rvert \leq 0.5$	127, $[0.00, +0.03]$	–	0 / 30
SPOT, maximum likelihood, declustered	123, $[-0.04, +0.02]$	21 / 24	20 / 20 (bursts: 10 / 20)
DSPOT, maximum-likelihood tail	127, $[0.00, +0.03]$	–	0 / 24

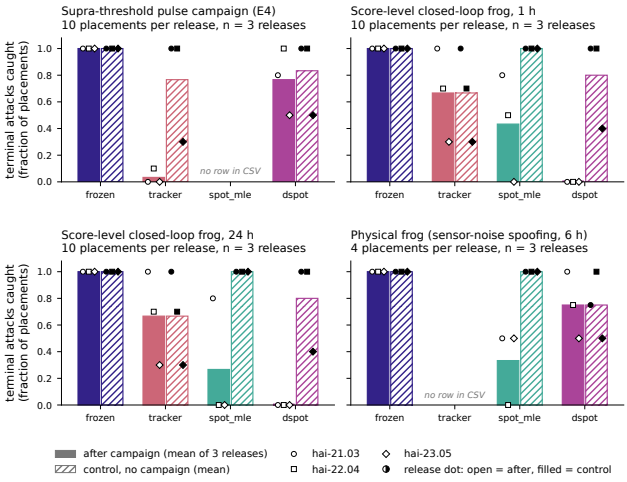


Figure 4: Terminal attacks caught after each campaign (solid) and in the matching control (hatched), for frozen, tracker, spot_mle and dspot. Each bar is the mean over the three spent HAI releases; dots are the per-release values, and every release had the same number of placements, so the mean equals the pooled fraction. Panels: the supra-threshold pulse campaign (E4, 10 placements per release); the score-level closed-loop frog at 1 h and 24 h (10 placements); and the physical frog, which spoofs noise on three sensors for 6 h (4 placements). Data status: all four panels are exploratory results on the normal data of the spent releases. The source CSV has no spot_mle row for E4 and no tracker row for the physical frog; both slots are left empty and labelled.

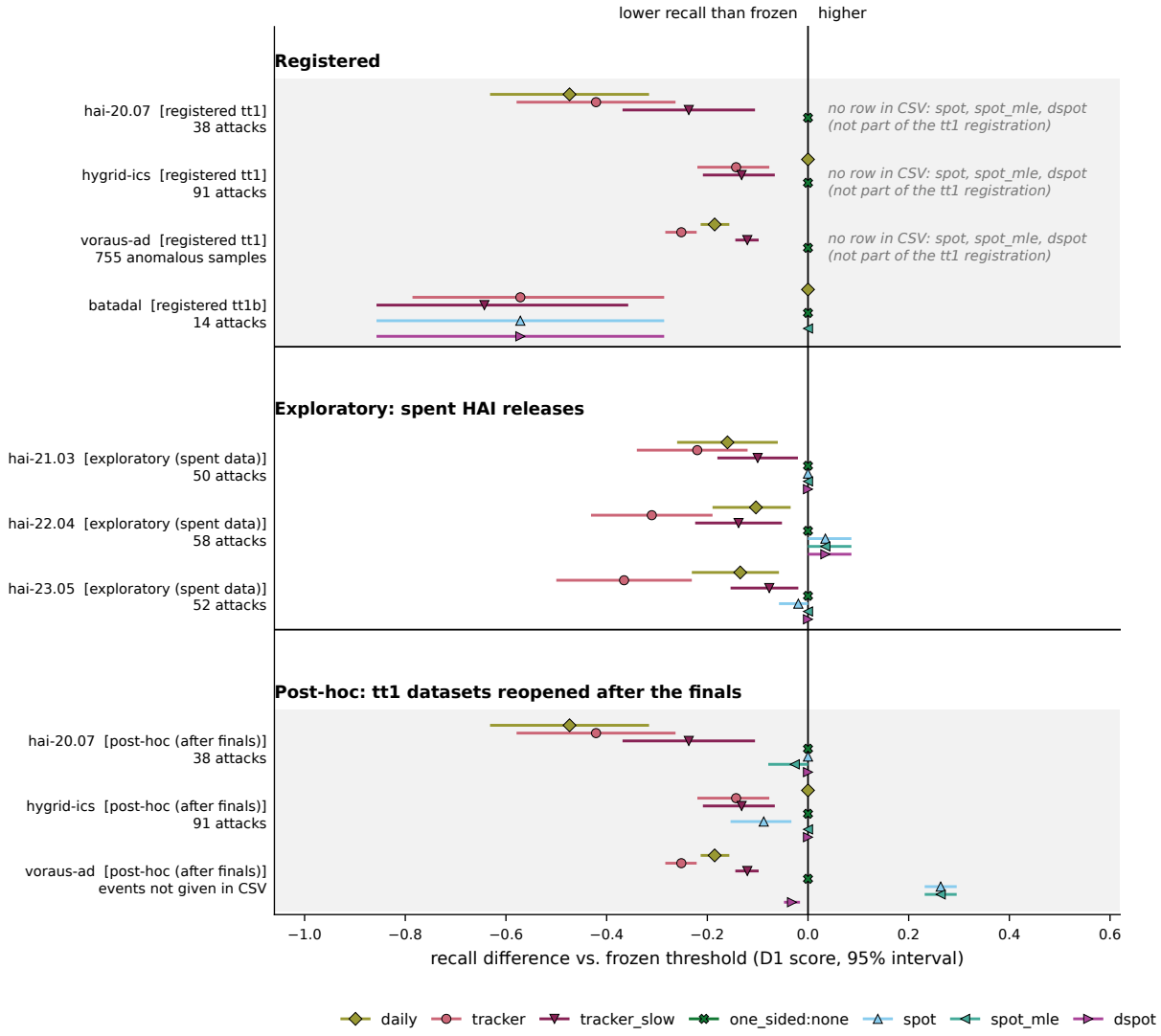


Figure 5: Paired difference in event recall between each adaptive rule and the frozen threshold on D1, with 95% intervals. Values left of 0 mean lower recall than frozen. Three blocks, each row labelled with its data status: (1) registered results from tt1 (hai-20.07, hygrid-ics, voraus-ad) and tt1b (batadal); (2) exploratory development results on the spent HAI releases; (3) post-hoc results on the tt1 datasets, reopened after the finals. spot, spot_mle and dspot have no rows for the registered tt1 datasets because those rules were not part of the tt1 registration; the gaps are labelled and the post-hoc block supplies the comparisons. one_sided:none equals frozen (0 $[0, 0]$) in every block. The positive spot and spot_mle values for voraus-ad (post-hoc) are not a gain: with 190 calibration samples SPOT ran at a per-sample false-positive rate of 0.32 against the 0.05 target (Section 7.1).

D How the hypothesis changed during development

All development used normal training data and the already-opened attack files of HAI 21.03, 22.04 and 23.05. We report it because the registered claim differs from the one we set out to test.

(1) *Tighten-only with automatic loosening.* Our starting design let a threshold tighten itself and loosen only through a statistical permit. A slow sub-threshold drift ("boiling frog" in the style of Rubinstein et al.) did not poison any rule: the world model absorbed the drift. Supra-threshold pulses did inflate the trackers, but every automatic permit we tried either opened under the pulses (spread permit: 3 of 30 terminal attacks caught in development round r3, 5 of 30 in round r4) or failed to follow benign drift (shift permit).

(2) *Static safety margin.* A margin at half the calibration exceedance rate cut the false-alarm rate on pseudo-deployment splits without always reaching the budget (HAI 20.07 development: 3.79 episodes per day against 4.42 for frozen, budget 2), and cost 4 to 10 points of recall on the spent attacks (interval lower bounds down to -0.17), so we registered a smaller margin and made it secondary.

(3) *Frozen versus adaptive.* On the spent attacks the frozen threshold held the budget on all three real test periods (0.66, 1.48 and 1.27 episodes per day), while the tracker lost 22, 31 and 37 points of recall with intervals entirely below zero. This became the registered primary.

E Every registered rule of tt1d

Table 17 gives the registered D1 outcomes of all twenty rules on both datasets. By attack type (21, 22, 23, 24; 34 / 75 / 41 / 95 events) the frozen threshold detected 34 / 63 / 28 / 95, the tracker 34 / 63 / 28 / 57, daily recalibration 34 / 57 / 23 / 51 and the capped tracker 34 / 71 / 34 / 95.

Table 17: tt1d, D1, every registered rule. Tennessee Eastman: first events detected (of 141), followers detected under the any-alarm and onset counts (of 104), and false-alarm episodes over the budget on the unlabelled time of the attack runs and on the sealed attack-free runs. ProvICS: events detected under both counts (of 32) and false alarms over the budget on the unlabelled time of the attack recording. Data status: registered (tt1d; ProvICS under amendment 1).

	Tennessee Eastman					ProvICS		
Rule	first	followers	onset	FA, attack runs	FA, sealed	any alarm	onset	FA
frozen	116	104	44	1.55	0.98	25	4	33.0
tracker	116	66	37	0.82	1.21	25	4	6.5
tracker_slow	110	88	50	0.74	0.83	25	4	18.8
tracker, $\eta = 0.01\,\tau_0$	123	55	37	1.02	1.64	24	6	1.9
daily	106	59	40	1.23	1.22	25	4	33.0
tighten-only	122	104	44	1.81	1.57	25	4	33.0
daily, episodes left out	113	104	44	1.87	1.33	25	4	33.0
tracker, hold-off	124	104	45	1.65	1.45	25	4	33.6
tracker, capped rise	130	104	44	2.35	2.12	25	4	37.5
SPOT (ML)	106	104	43	1.41	0.84	25	4	35.6
static $1.1\,\tau_0$	80	104	45	0.83	0.32	25	4	37.5
static $1.25\,\tau_0$	71	104	43	0.50	0.16	25	4	25.2
static $1.5\,\tau_0$	71	104	44	0.25	0.11	25	4	5.8
tracker, ceiling 1.1	123	104	43	1.82	1.51	25	4	37.5
tracker, ceiling 1.25	121	104	41	1.26	1.30	25	4	25.2
tracker, ceiling 1.5	116	104	42	0.94	1.21	25	4	11.0
daily, ceiling 1.25	106	104	44	1.45	1.22	25	4	33.0
SPOT, ceiling 1.25	106	104	43	1.41	0.84	25	4	35.6
SPOT, approval every 6 h	106	104	43	1.42	0.84	25	4	31.7
SPOT, approval daily	106	104	43	1.42	0.84	25	4	31.7